

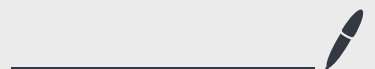
CSI 2132

Databases 1

Deliverable 1

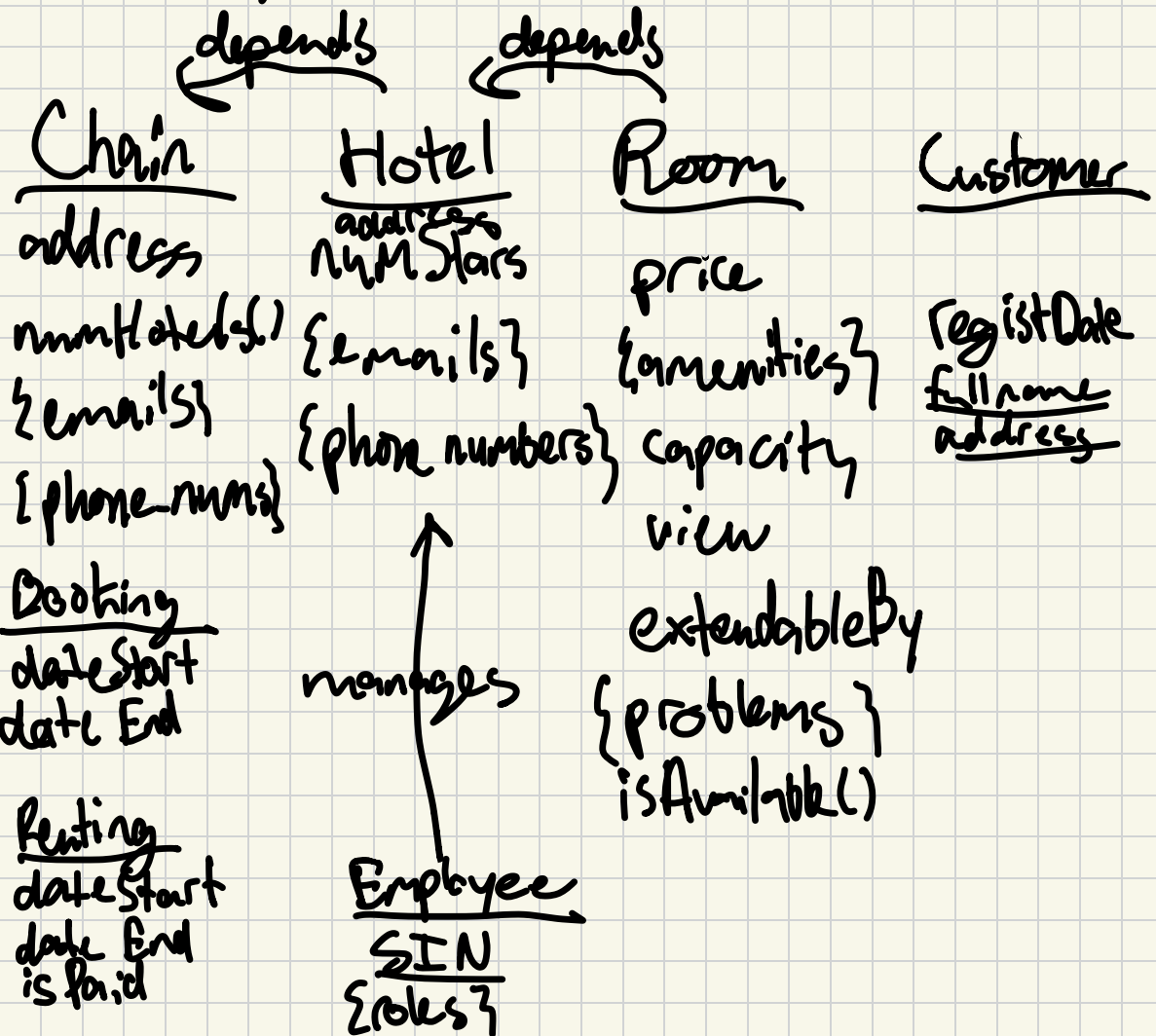
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Brainstorm

entities: chain, hotel, room, customer,
employee, booking?



constraints brainstorm:

Stars (1-5)

room.
Capacity (int)

room.view {sea, mountain, none}

Room.isAvailable() false while Booking or Renting

Reservation customer and room FK optional

hotel cannot exist without employee manager

manages relationship restricted to employee roles

price > 0

= manager

start Date < end Date

Reservation fails if Room.isAvailable == false

Assumptions:

Hotel is weak entity set to HotelChain

Room is weak entity set to hotel (∴ also HotelChain)

Room needs id (ex; room number) PK alongside aggregation PK

generalize Booking and Renting to Reservation

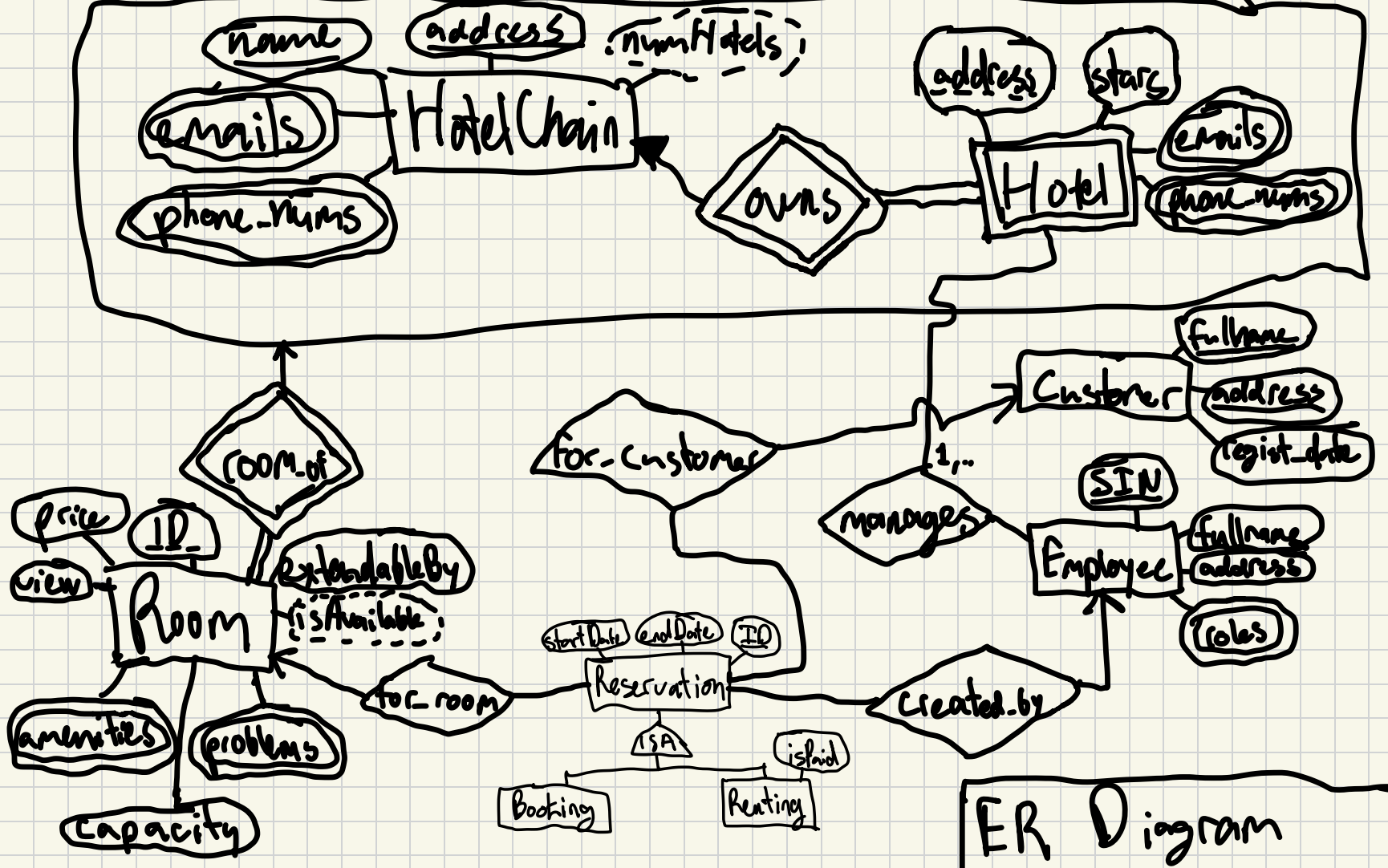
Reservation must have unique ID (as it can exist w/o FK for archive)

Reservation is either Booking or Renting

Employee manually creates new Renting when customer arrives.

Employee manage Hotel is N-N relationship as Hotel can have multiple managers. And 1 employee can manage multiple hotels (ex; regional manager)

∴ aggregation on HotelChain + Hotel



Relational Model

- = relational schema
- = PK and FK explanation
- = constraints

HotelChain(name, address, numHotels())

PK unique name/address combination

numHotels() derived off number of hotels with FK referencing this HotelChain

HC_phones(hc_name, hc_address, phone_num)

HC_emails(hc_name, hc_address, email)

FK hc_name references HotelChain.name

FK hc_address references HotelChain.address

PK unique combination of FK with attribute (email or phone num)

Hotel(hc_name, hc_address, address, stars)

FK hc_name references HotelChain.name

FK hc_address references HotelChain.address

PK unique combination of FKs and address

stars of type $\text{int} \in [1, 5]$

Hotel must be created alongside an initial HotelManager (fails otherwise)

H_phones(h_name, h_address, phone_num)

H_emails(h_name, h_address, email)

FK h_name references Hotel.name

FK h_address references Hotel.address

PK unique combination of FK with attribute (email or phone num)

Room(ID, h-address, hc-address, hc-name, price, extendableBy, view, isAvailable, capacity)

FK hc-name references HotelChain.name

FK hc-address references HotelChain.address

FK h-address references Hotel.address

PK unique combination of all FK with ID

price float ≥ 0.00

capacity int ≥ 1

extendableBy int 0 by default

view string $\in \{ "sea", "mountain", "none" \}$

isAvailable bool derived from if there's an ongoing Reservation for this room.

RoomAmenity (r_id, h_addr, hc_addr, hc_name, amenity_desc)

FK: {r_id from Room.ID, h_addr from Hotel.address

hc_addr from HotelChain.address, hc_name from HotelChain.name}

PK unique combination of all FK and amenity_desc

amenity_desc of type string

RoomProblem (r_id, h_addr, hc_addr, hc_name, problem_desc)

FK: {r_id from Room.ID, h_addr from Hotel.address

hc_addr from HotelChain.address, hc_name from HotelChain.name}

PK unique combination of all FK and problem_desc
problem_desc of type string

Customer(fullname, address, registr_date)

PK unique fullname and address combination

Employee(SIN, fullname, address)

PK unique SIN

EmployeeRole(e_sin, role_name)

FK e_sin references Employee.SIN

PK unique combination of FK and role_name

role_name of type string and is mandatory

HotelManager(e_sin, h_addr, hc_name, hc_addr)

this comes from
N-N "manages" relationship

FK e_sin to Employee.SIN

FK h_addr to Hotel.address

FK hc_addr to HotelChain.address

FK hc_name to HotelChain.name

PK unique combination of all FK

creation of HotelManager will fail if no EmployeeRole corresponding to current employee with role_name containing the substring "manager" (ex, "regional manager" is valid)

Booking(ID, start Date, end Date,
r_id, h_addr, hc_addr, hc_name,
c_name, c_addr,
created-by)

all 3 of these lines
are optional FK

PK is unique ID (for archiving)

FK: { r_id from Room.ID, h_addr from Hotel.address

hc_addr from HotelChain.address, hc_name from HotelChain.name
c_name from Customer.name, c_addr from Customer.address
created-by from Employee.SIN }

Start Date < end Date

Booking creation fails if target Room.isAvailable == false

Renting(ID, start+Date, endDate, isPaid

r_id, h-addr, hc-addr, hc-name,
C-name, C-addr,
created-by)

all 3 of these lines
are optional FK

PK is unique ID (for archiving)

FK: { r_id from Room.ID, h-addr from Hotel.address

hc-addr from HotelChain.address, hc-name from HotelChain.name
C-name from Customer.name, C-addr from Customer.address
Created-by from Employee.SIN }

Start Date < end Date

Renting creation fails if target Room.isAvailable == false